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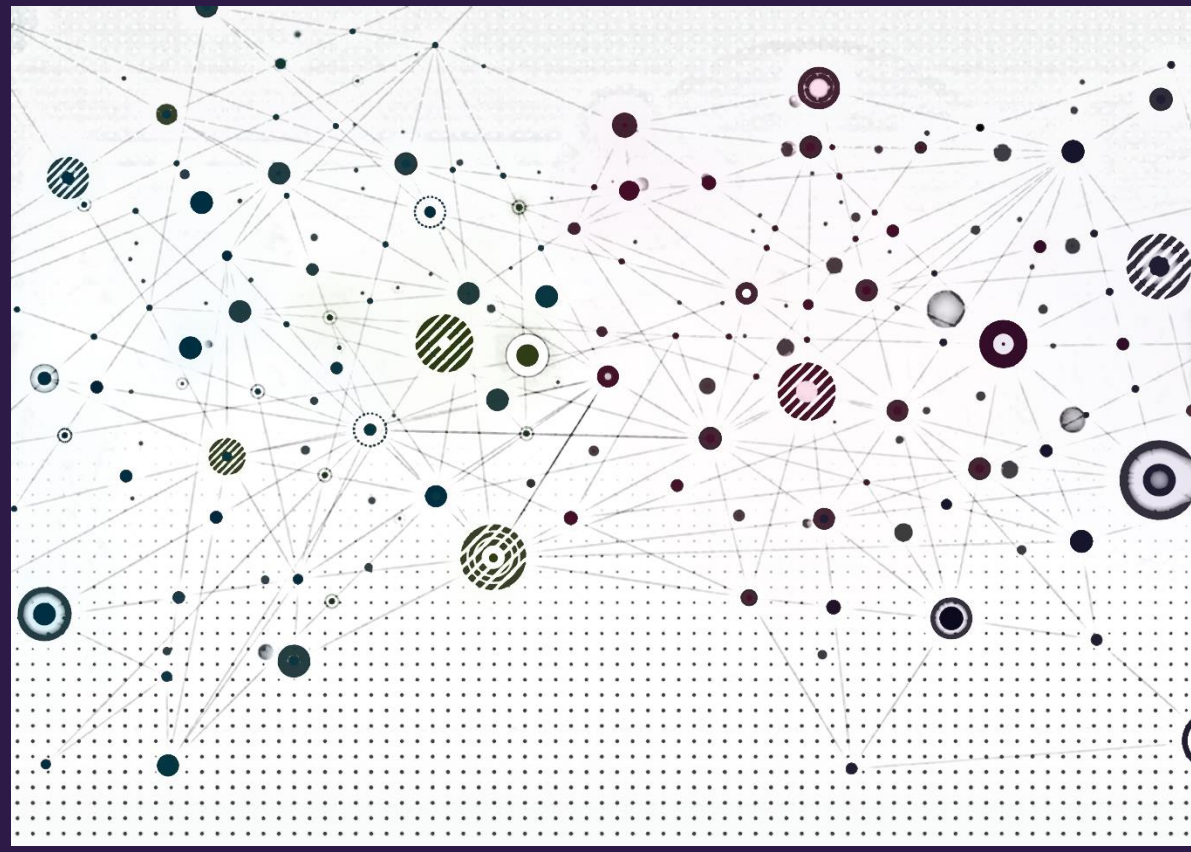
A Generalized Perspective on GNNs



Overview

- Recap
- Message passing framework
- Standard GNNs in the message passing framework
- Applicable deep learning tools
- Stacking GNN layers
- Graph manipulations

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Recap

Deep learning on graphs

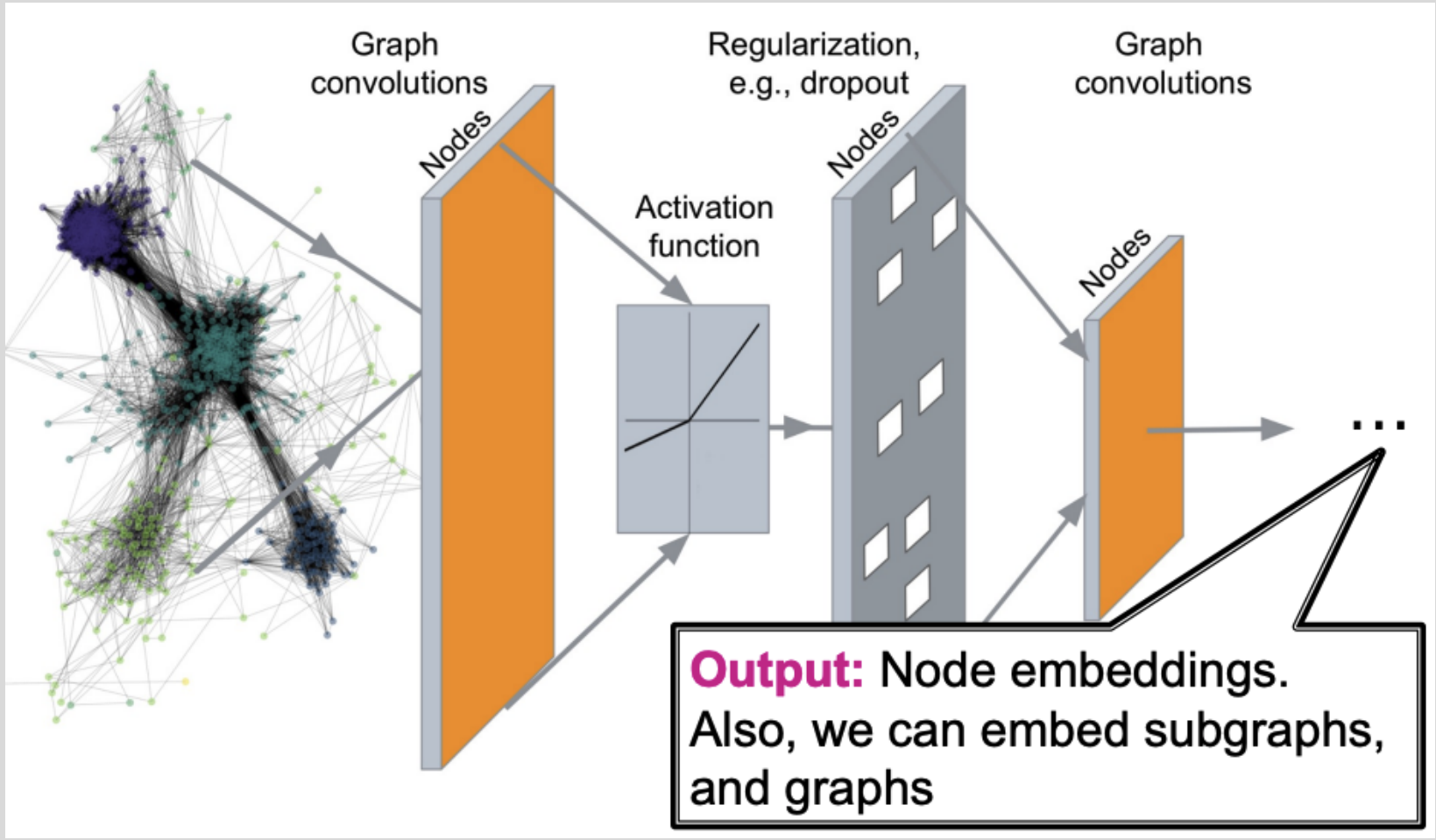


Image credit: Jure Leskovec, http://cs224w.stanford.edu

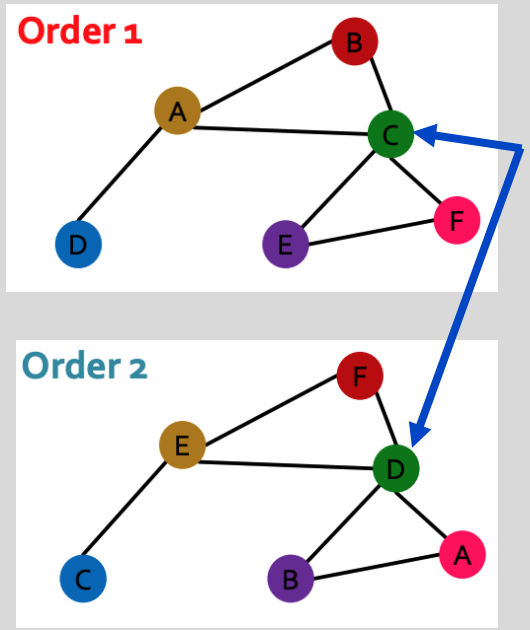
Graph deep learning paradigm

- Assume we have a graph $\boldsymbol{G}$
- V is the set of vertices in $\boldsymbol{G}$
- $\mathbf{A}$ is the adjacency matrix (assume unweighted, undirected for now)
- $\mathbf{X} \in \mathbb{R}^{|V| \times d_l}$ is a matrix of **node features (this is new!)**
  - One d_l-dimensional feature vector per node
  - Examples of node features
    - User profile data, user image
    - Gene expression, gene functional properties
  - What if my nodes don't have features?
    - One-hot embeddings - not recommended as this makes the model transductive
    - Constant vector of 1 – model is again inductive, features will be updated (more to come)

Permutation invariance and equivariance

- Permutation **invariant** – permute the input, the output is unchanged

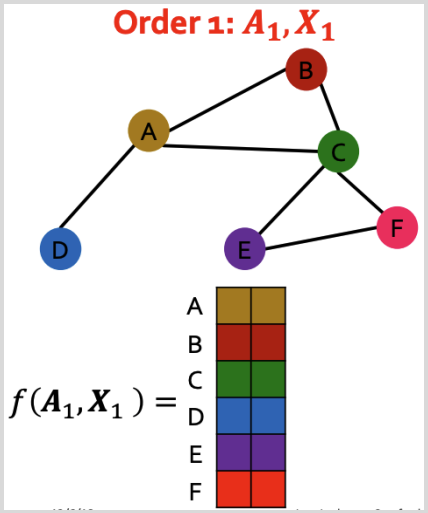
$$f(\mathbf{A}, \boldsymbol{X}) = f\left(\mathbf{P}\mathbf{A}\boldsymbol{P}^{\boldsymbol{T}}, \boldsymbol{P}\boldsymbol{X}\right)$$

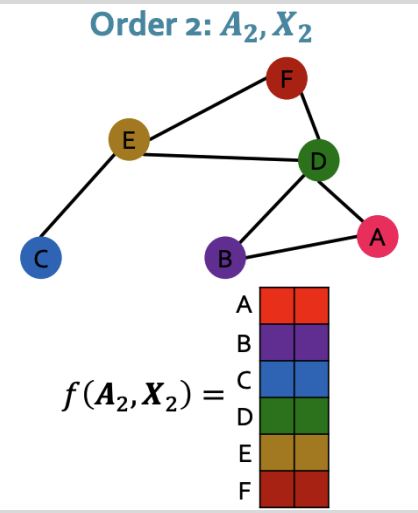


Permutation invariance holds for node embeddings (can also hold for graph embeddings by aggregating over all node embeddings)

- Permutation **equivariance** - permute the input, the output is permuted in the same way

$$\mathbf{P}f(\mathbf{A}, \boldsymbol{X}) = f\left(\mathbf{P}\mathbf{A}\boldsymbol{P}^{\boldsymbol{T}}, \boldsymbol{P}\boldsymbol{X}\right)$$

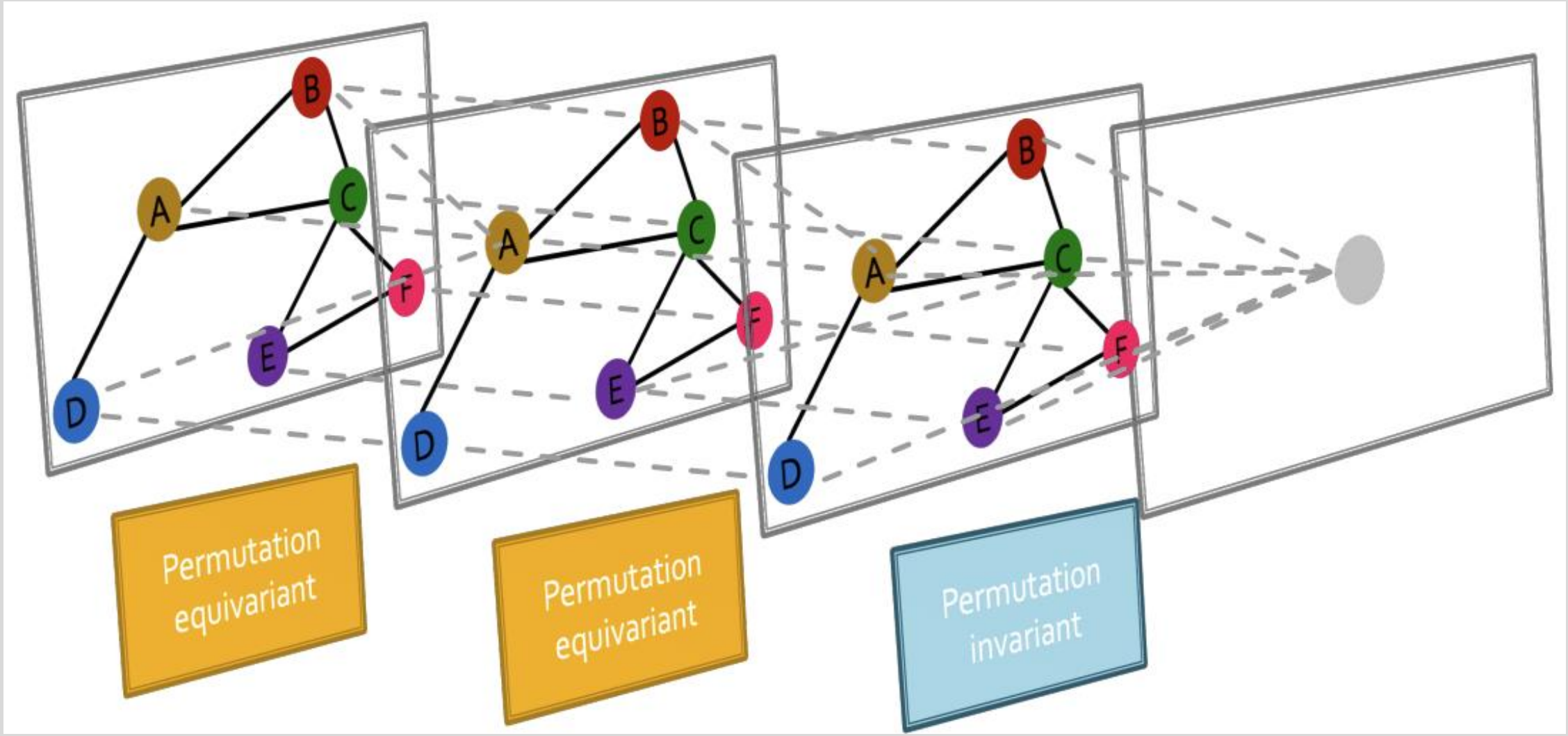




Permutation equivariance is property of the output embedding matrix

Image credit: Jure Leskovec, http://cs224w.stanford.edu

GNNs chain permutation invariant and equivariant operations



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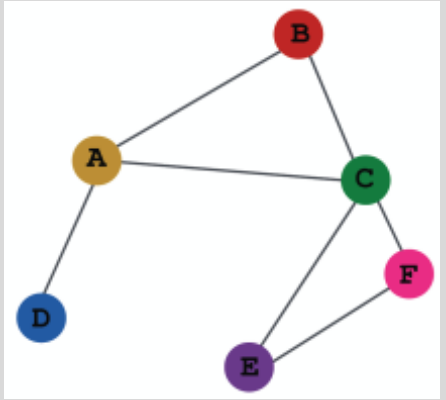
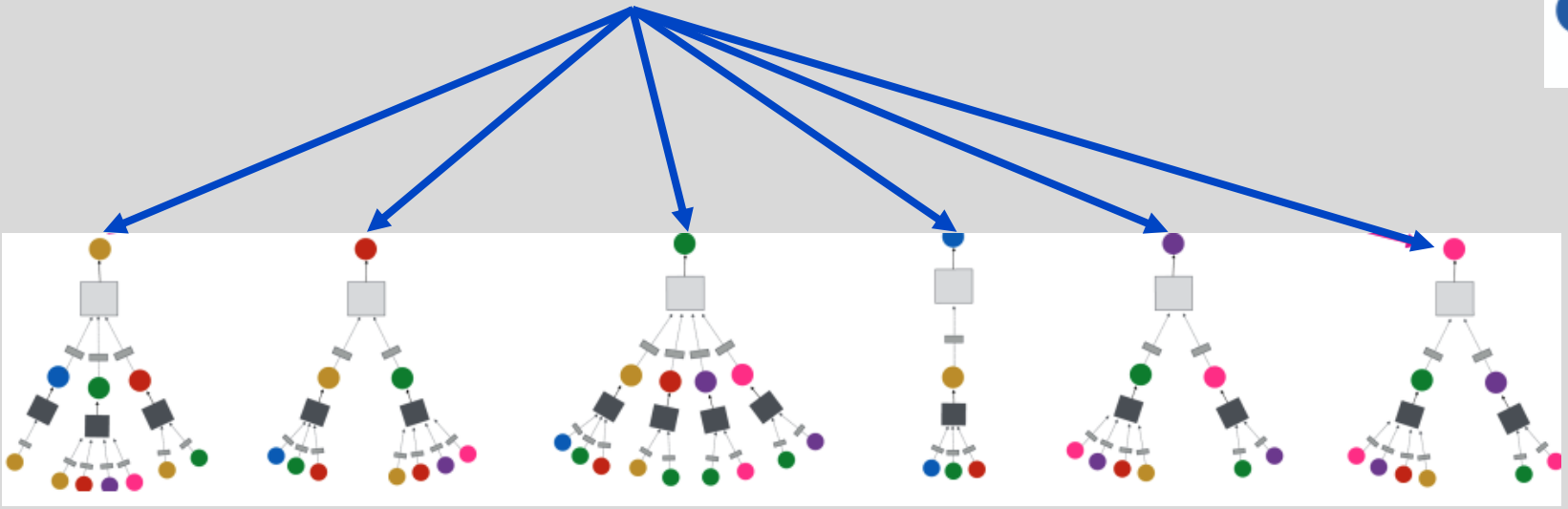
Neighbor node aggregation

- Every neighborhood defines a computation graph

Input graph



Each node has a unique computation graph based on its neighborhood



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The math of graph convolution networks in matrix form

- From the previous result, we can rewrite the update function over all nodes as:

$$\mathbf{H}^{(k+1)} = \sigma\left(\tilde{\mathbf{A}}\mathbf{H}^{(k)}\mathbf{W}^{(k)^T}\right)$$

where $\tilde{\mathbf{A}} = \mathbf{D}^{-1}\mathbf{A}$

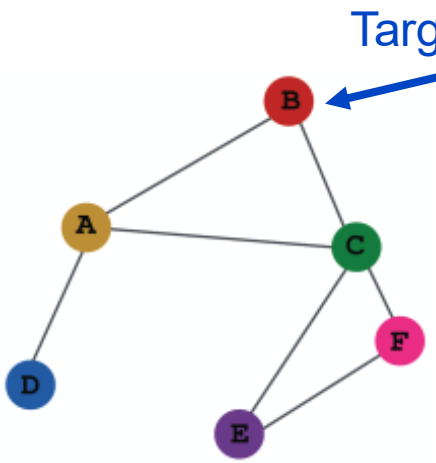
normalization

neighborhood aggregation

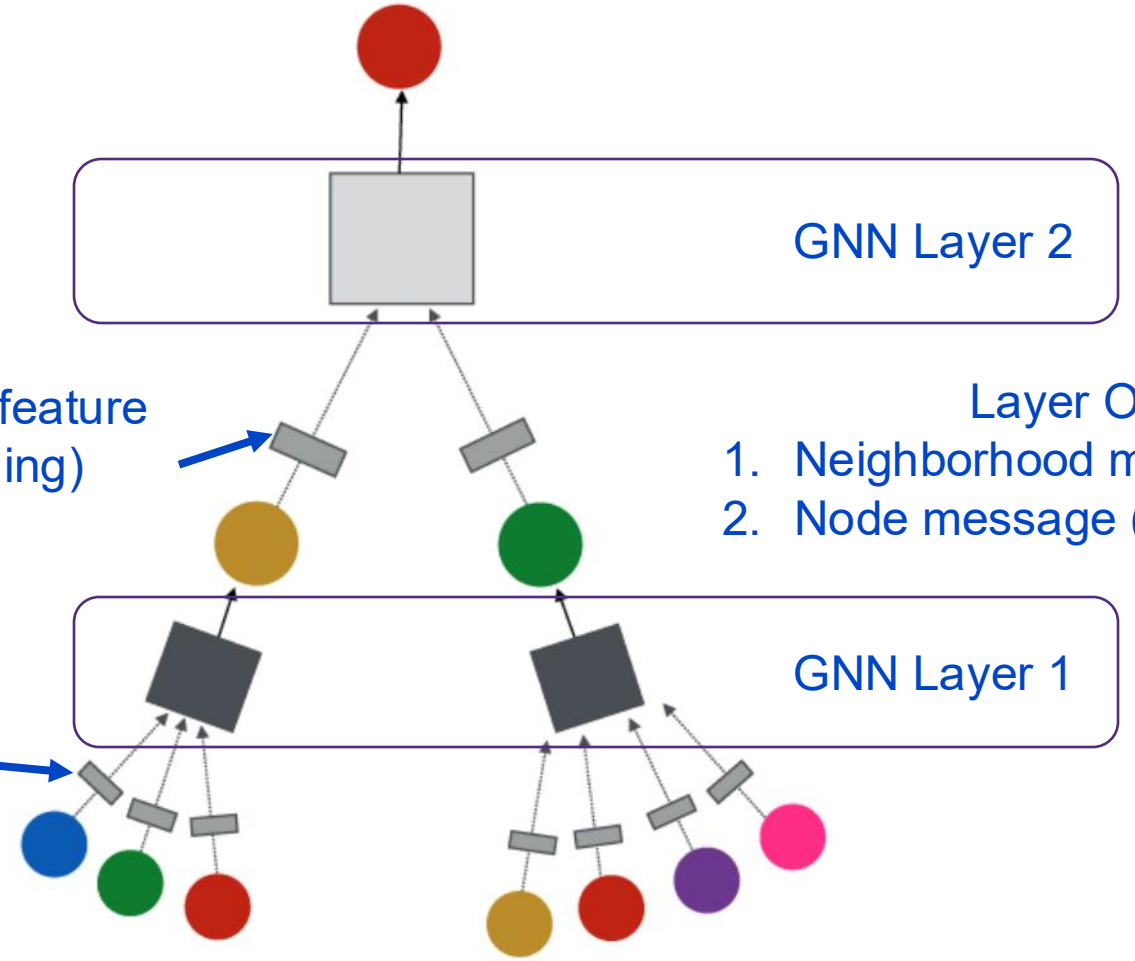
Can we generalize this approach?

Message passing framework

Message passing in GNNs



Input graph



Message: Updated feature vectors (embedding)
$\mathbf{h}_v^1$

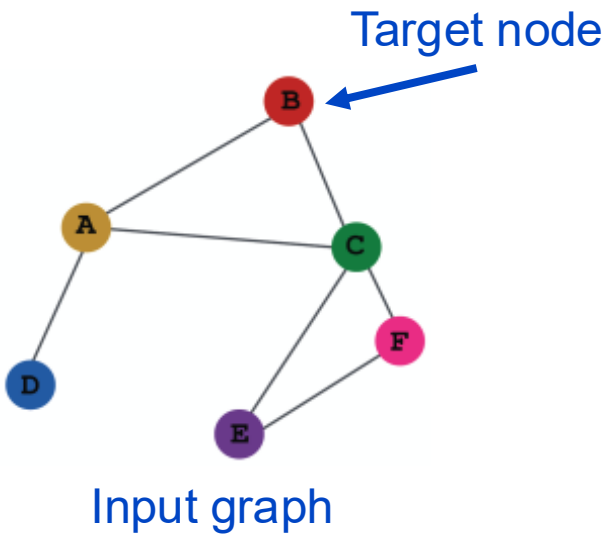
Message: Input feature vectors
$\mathbf{h}_v^0 = \mathbf{x}_v$

Layer Operations

1. Neighborhood message aggregation
2. Node message (embedding) update

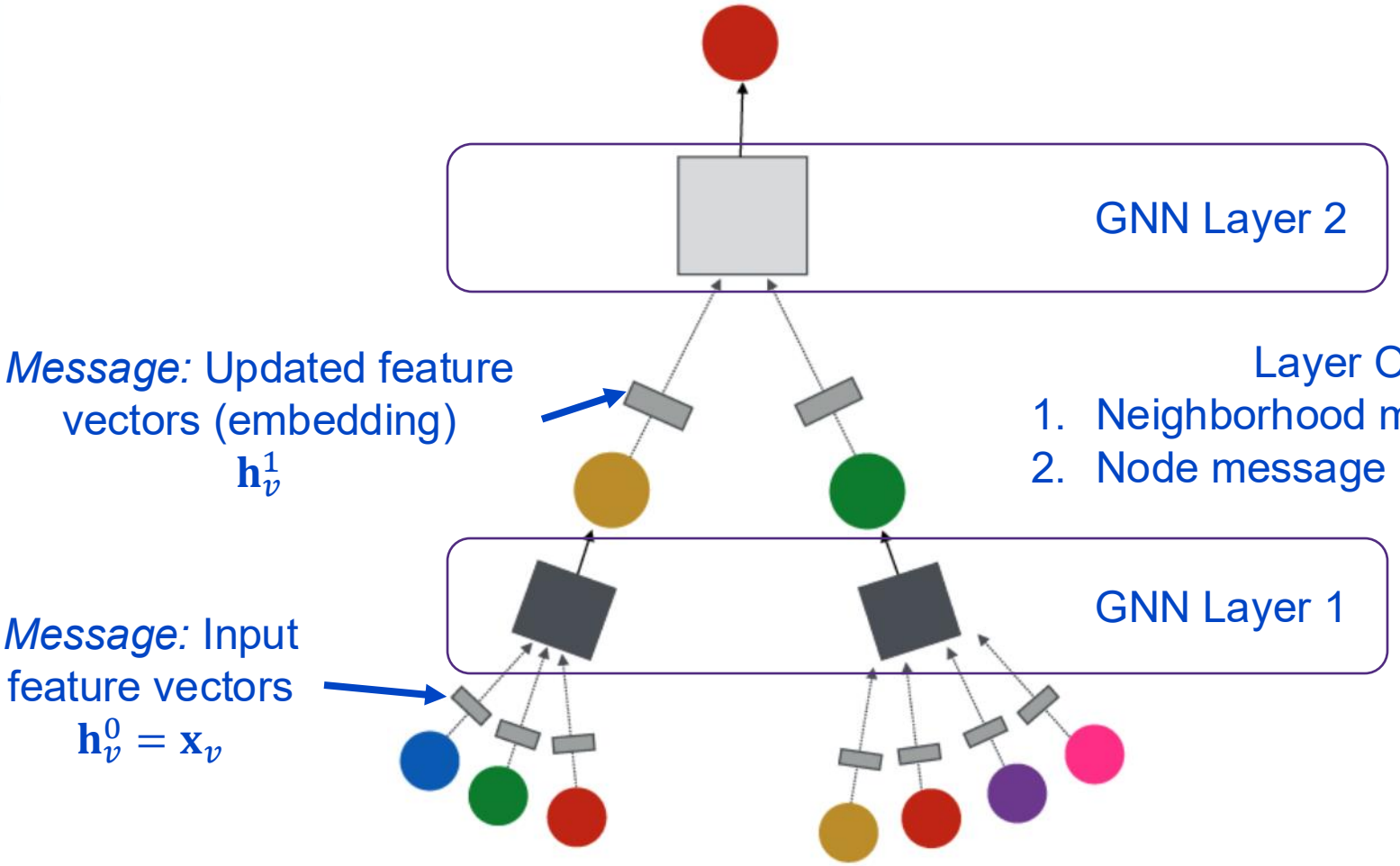
Image credit: Jure Leskovec, http://cs224w.stanford.edu

Message passing in GNNs



GNN Layer ≡ Aggregation → Update

- Different architectures under this perspective
- GCN, GraphSage, GAT, …



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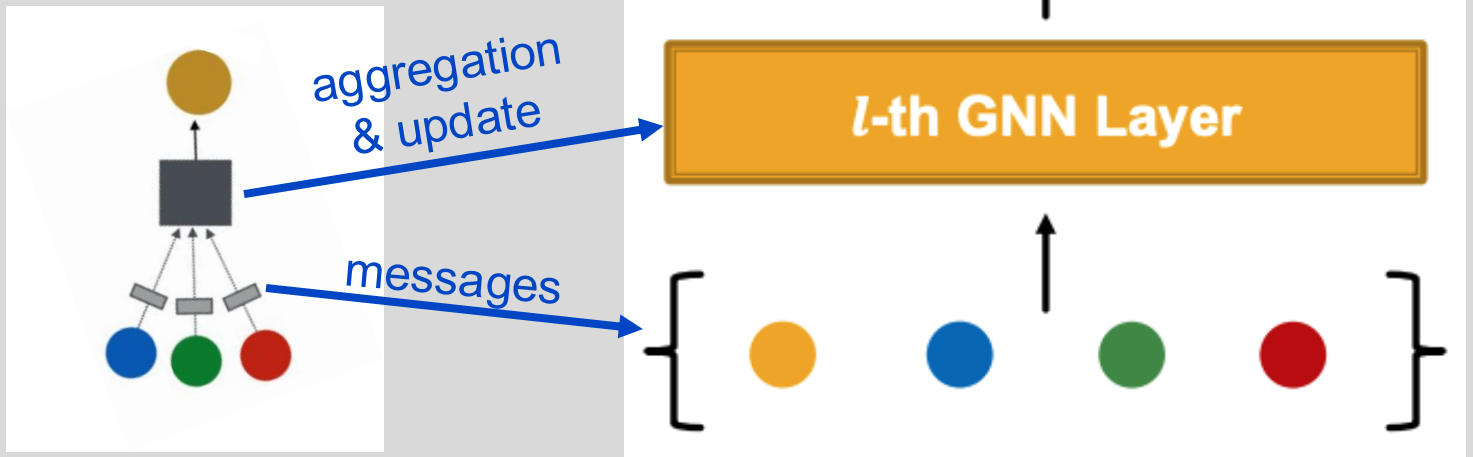
GNN layer concept

- A single GNN layer
  - **Main idea**: Convert a set of input vectors into a single vector while maintaining permutation invariance (individual node embeddings) equivariance (output matrix of all node embeddings)
  - Two step process:
    1. Node message aggregation
    2. Node embedding update



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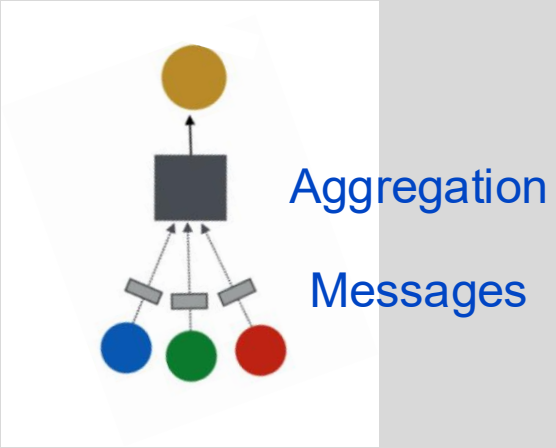
Node neighborhood message aggregation

- Node v will aggregate messages from its neighbors $u \in N(v)$ to form a single neighborhood message:

$$\mathbf{m}_{N(v)}^{(k)} = Agg^{(k)}\left(\left\{\mathbf{h}_u^{(k)}, \forall u \in N(v)\right\}\right)$$

  where $Agg^{(k)}$ is any differentiable, permutation invariant function
- Examples: Sum, mean, max (element wise operations)



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Node message (embedding) update

- After neighborhood aggregation, node v will update its embedding:

$$\mathbf{h}_v^{(k+1)} = Update^{(k)}\left(\mathbf{h}_v^{(k)}, \mathbf{m}_{N(v)}^{(k)}\right)$$

where $Update^{(k)}$ is any differentiable function (typically a neural network)

- Example: fully connected linear layer $\mathbf{h}_v^{(k+1)} = \left[\mathbf{h}_v^{(k)} | \mathbf{m}_{N(v)}^{(k)}\right] \mathbf{W}^{(k)^T}$

update matrix (linear layer)

vector concatenation

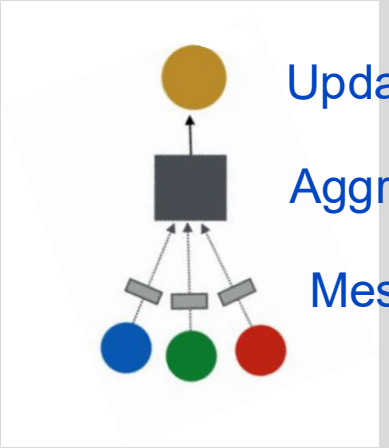


Image credit: Jure Leskovec, http://cs224w.stanford.edu

Message aggregation and update as a GNN layer

- Combine the neighborhood message aggregation and node embedding update to form a single GNN layer
  1. Aggregation: each node aggregates messages (embeddings) of neighbor nodes

$$\mathbf{m}_{N(v)}^{(k)} = Agg^{(k)}\left(\left\{\mathbf{h}_u^{(k)}, \forall u \in N(v)\right\}\right)$$

  2. Update: each node updates its embedding (message for next layer)

$$\mathbf{h}_v^{(k+1)} = Update^{(k)}\left(\mathbf{h}_v^{(k)}, \mathbf{m}_{N(v)}^{(k)}\right)$$

  - The update usually includes a nonlinearity (activation), $\sigma(\cdot)$, such as ReLU, sigmoid

Nonlinear activation function review

- As with standard deep-learning methods, nonlinear activation functions are applied element wise to update a node's embedding
- Sigmoid – can be useful to restrict range of embedding values, though not the typical choice

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

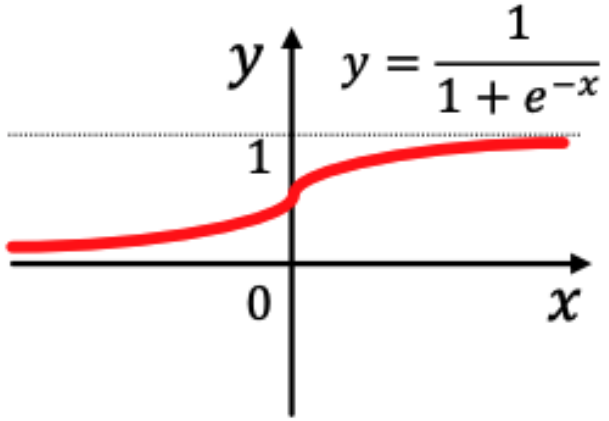
- ReLU – typical default

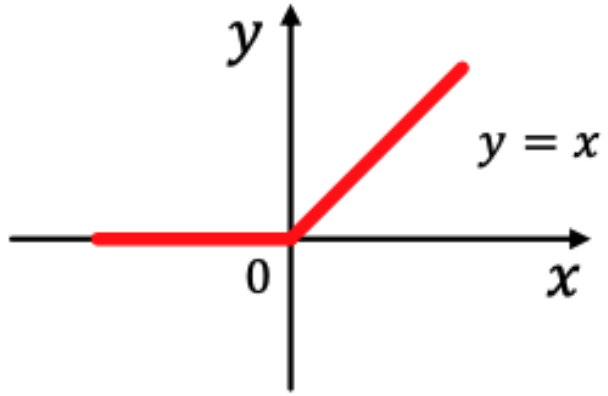
$$ReLU(z) = max(z, 0)$$

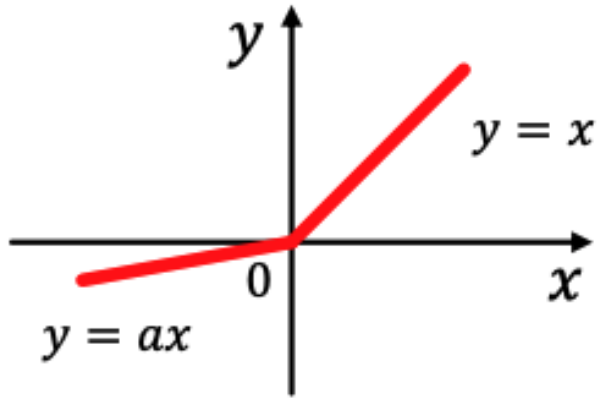
- Parametric ReLU

$$PReLU(z_i|a_i) = max(z_i, 0) + a_i min(z_i, 0)$$

  - a_i is a trainable parameter
  - empirical evidence suggests better performance than ReLU







GNNs in the message passing framework

Graph convolutional networks

- A GCN can be written as

$$\mathbf{h}_v^{(k+1)} = \sigma\left(\mathbf{W}^{(k)} \sum_{u \in N(v) \cup \{v\}} \frac{\mathbf{h}_u^{(k)}}{|N(v)|}\right)$$

Recall: Graph augmented with self loops

- How to write this as aggregation → update?

$$\mathbf{m}_{N(v)}^{(k)} = \sum_{u \in N(v) \cup \{v\}} \frac{\mathbf{h}_u^{(k)}}{|N(v)|}$$

$$\mathbf{h}_v^{(k+1)} = \sigma\left(\mathbf{W}^{(k)} \mathbf{m}_{N(v)}^{(k)}\right)$$

Graph convolution networks in matrix form

- The GCN can be written in matrix form as

$$\mathbf{H}^{(k+1)} = \sigma\left(\tilde{\mathbf{A}}\mathbf{H}^{(k)}\mathbf{W}^{(k)^T}\right)$$

  where $\tilde{\mathbf{A}} = \mathbf{D}^{-1}\mathbf{A}$

- How to write this as aggregation → update?

$$\mathbf{M}^{(k)} = \tilde{\mathbf{A}}\mathbf{H}^{(k)}$$

$$\mathbf{H}^{(k+1)} = \sigma\left(\mathbf{M}^{(k)}\mathbf{W}^{(k)^T}\right)$$

  where $\mathbf{M}^k \in \mathbb{R}^{|V| \times \boldsymbol{d}_k}$ is the matrix of message aggregation function outputs for all nodes (i.e., vector at row i is the aggregated message of the neighbors of node i (from the adjacency matrix ordering)

GraphSAGE (SAmple and aggreGatE) networks

- Message aggregation over all neighbors can be computationally expensive for large / dense graphs or high-degree nodes
- Instead , randomly sample a fixed sized subset of $N(v)$ for every new aggregation

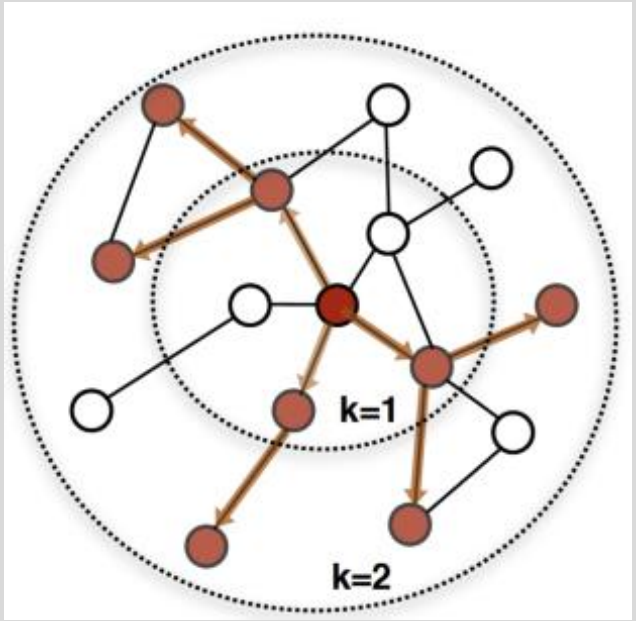


Image credit: Hamilton, Ying, Leskovec 2018

GraphSAGE (SAmple and aggreGatE) sampling

- For example, sample two (2) neighbors for message aggregation
  - Here, only nodes B and D pass messages to A

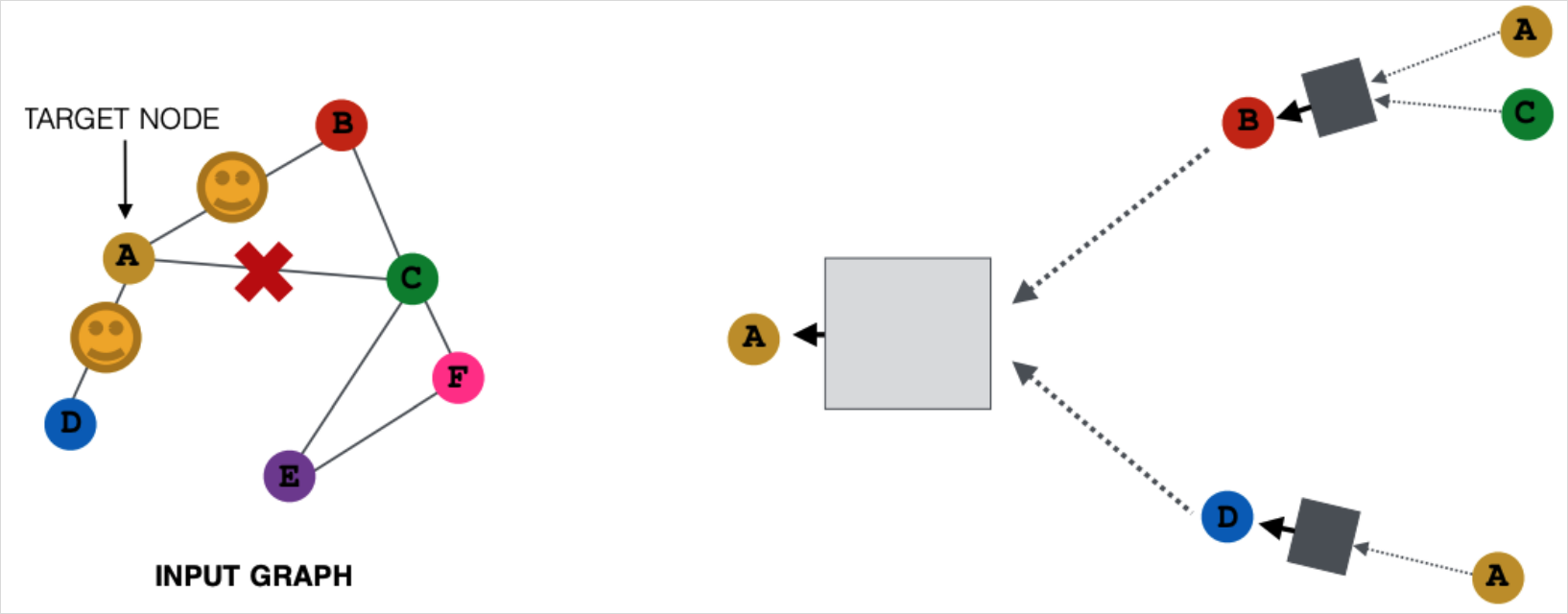
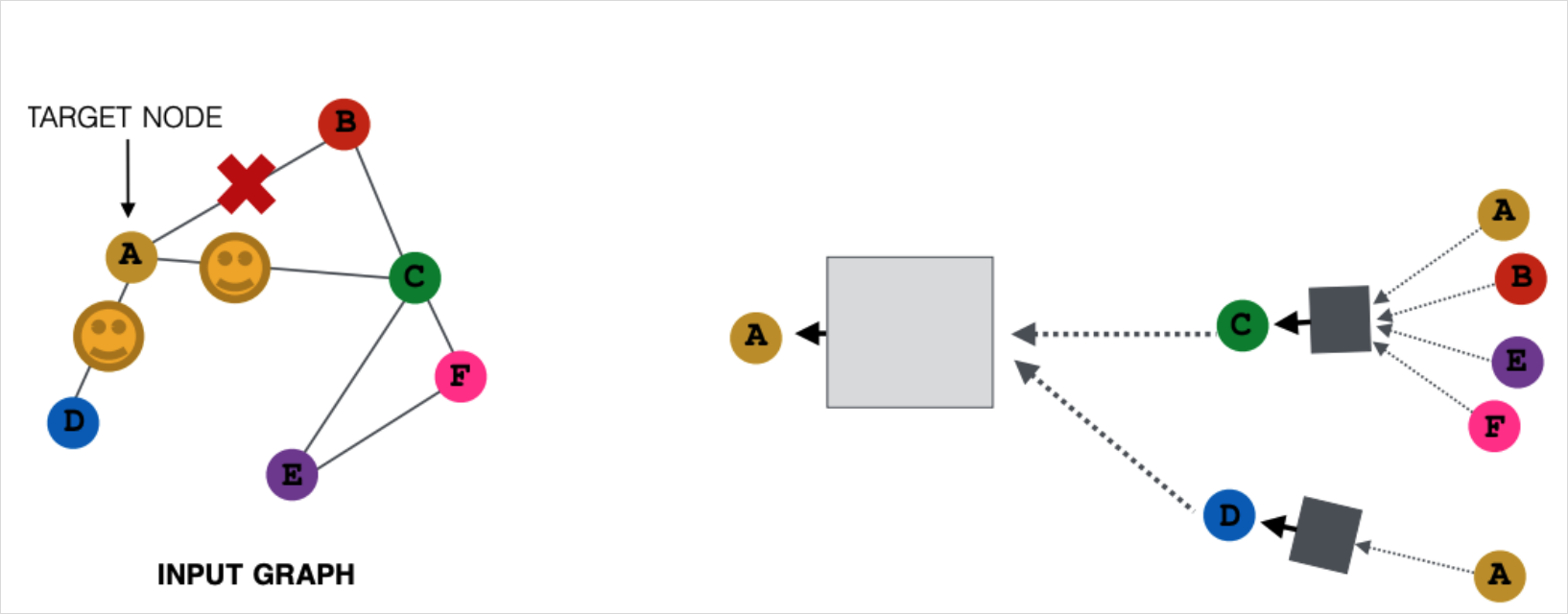


Image credit: Jure Leskovec, http://cs224w.stanford.edu

GraphSAGE (SAmple and aggreGatE) sampling

- For the next embedding update (layer, training batch, new sample), we can sample different neighbors
  - Here, only nodes C and D pass messages to A



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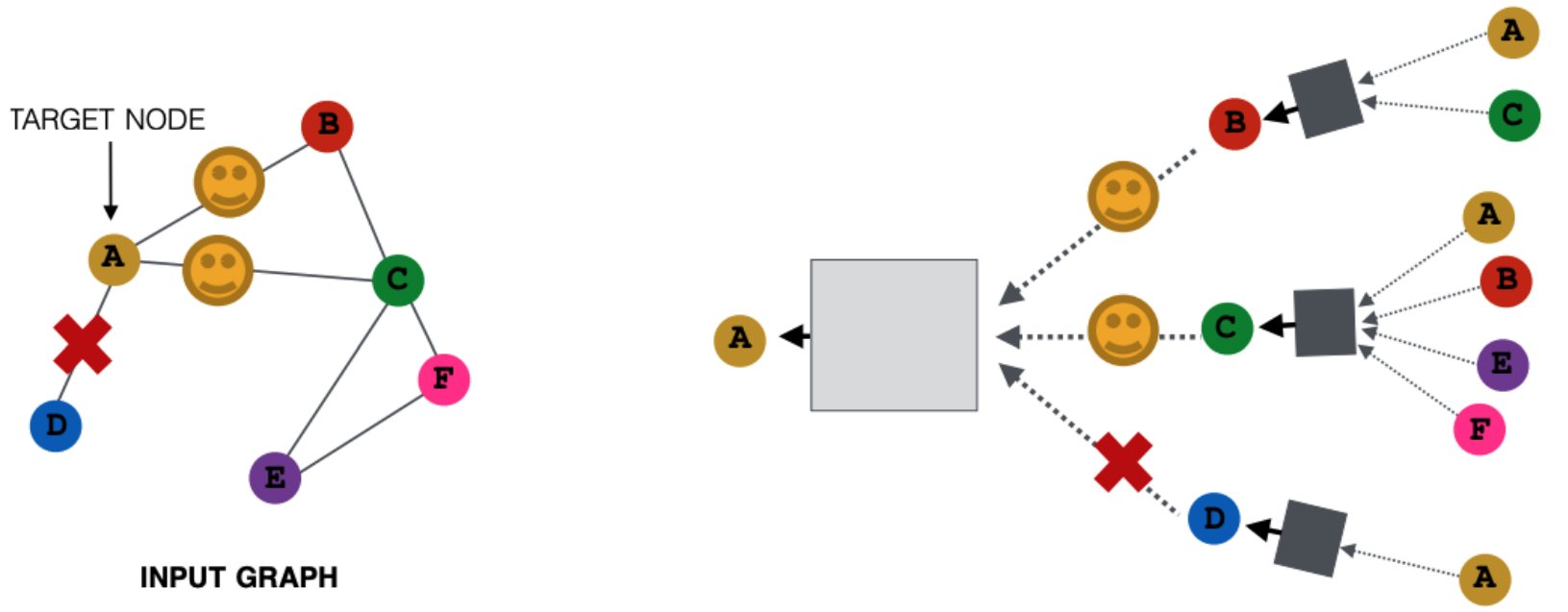
GraphSAGE (SAmple and aggreGatE) sampling

- In expectation (on average), we get embeddings similar to aggregating over all neighbors
  - Greatly reduces computational cost



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GraphSAGE node message aggregation

- Aggregate the messages from the sampled subset of $N(v)$ using permutation invariant function
- **Weighted average:** as in GCN

$$\mathbf{m}_{N(v)}^{(k)} = \sum_{u \in N(v)} \frac{\mathbf{h}_u^{(k)}}{|N(v)|}$$

- **Pool:** Apply an *MLP* to each node (same MLP on every node) and then apply an element wise vector function, $g(\mathbf{x}) \in \mathbb{R}^d \rightarrow \mathbb{R}^{\tilde{d}}$, such mean or max to the set of transformed embeddings

$$\mathbf{m}_{N(v)}^{(k)} = g\left[\left\{MLP\left(\mathbf{h}_u^{(k)}\right), \forall u \in N(v)\right\}\right]$$

GraphSAGE node message aggregation

- Janossy pooling: apply a permutation-sensitive function and average over many permutations
- Let $\pi_i \in \Pi$ denote a permutation function that maps the set $\{\mathbf{h}_u, \forall u \in N(v)\}$ to the sequence $\left(\tilde{\mathbf{h}}_1, \cdots, \tilde{\mathbf{h}}_{|N(v)|}\right)$ (here, $N(v)$ is the randomly sampled subset of neighbors)
- Consider a random sampling of permutation orders, Π
- Aggregate node messages as

ρ_Φ is a **permutation-sensitive** function, usually an LSTM

$$\mathbf{m}_{N(v)}^{(k)} = MLP_\Theta\left(\frac{1}{|\Pi|}\sum_{\pi\in\Pi}\rho_\Phi[\pi_i(\{\mathbf{h}_u, \forall u \in N(v)\})]\right)$$

sum over multiple permutations

GraphSAGE embedding update

- Regardless of node message aggregation, the same update is applied

$$\mathbf{h}_v^{(k+1)} = \sigma\left(\mathbf{W}^{(k)}\left[\mathbf{h}_v^{(k)} \mid \mathbf{m}_{N(v)}^{(k)}\right]\right)$$

vector concatenation with node's previous layer embedding

- (Optional) ℓ_2 normalization at every layer

$$\mathbf{h}_v^{(k)} \leftarrow \frac{\mathbf{h}_v^{(k)}}{\left\|\mathbf{h}_v^{(k)}\right\|_2}$$

  - Puts all embeddings on the same scale (same ℓ_2 norm)
  - Tends to improve performance (though not always)

Graph attention networks

- Idea: weight neighbor nodes differently based on some importance function

$$\mathbf{m}_{N(v)}^{(k)} = Agg^{(k)}\left(\left\{\alpha_{vu}^{(k)}\mathbf{h}_u^{(k)}, \forall u \in N(v) \cup v\right\}\right)$$

  where $\boldsymbol{\alpha_{vu}^{(k)}}$ **is the attention weight** placed on $\mathbf{h}_u^{(k)}$
- In GCN / GraphSage
  - $\alpha_{vu}^{(k)} = \frac{1}{|N(v)|}\ \forall\ k$ provides equal attention to all neighbor nodes for a given node, v, i.e., all nodes are equally important to v
  - However, governed by graph structure, specifically $deg(v)$ (node degree), i.e.

$$\alpha_{v_1u}^{(k)} = \frac{1}{deg(v_1)} \neq \alpha_{v_2u}^{(k)} = \frac{1}{deg(v_2)}$$

Graph attention networks

- Idea: weight neighbor nodes differently based on some importance function

$$\mathbf{m}_{N(v)}^{(k)} = Agg^{(k)}\left(\left\{\alpha_{vu}^{(k)}\mathbf{h}_u^{(k)}, \forall u \in N(v) \cup v\right\}\right)$$

  where $\boldsymbol{\alpha_{vu}^{(k)}}$ **is the attention weight** placed on $\mathbf{h}_u^{(k)}$

- What if neighbor nodes are not equally important?
  - Attention, $\alpha_{vu}^{(k)}$, should focus on the important parts of the input data and discard the rest
  - Can we learn a function that selects the important parts of the input?

Graph attention networks

- **Goal:** Learn attention weights $\alpha_{vu}^{(k)}$ that place optimal importance on node neighbors for each graph node
- **Idea:** Compute node embeddings, $\mathbf{h}_v^{(k)}$, using an **attention strategy:**
  - Nodes should attend over their neighborhoods' messages
  - Implicit attention weight specification (i.e., formulate this in a learnable manner)

Graph attention networks message aggregation + attention

- Formulate attention in the message passing framework

$$\mathbf{m}_{N(v)}^{(k)} = \sum_{u \in N(v)} \alpha_{v,u}^{(k)} \mathbf{h}_u^{(k)}$$

where $\alpha_{v,u}^{(k)}$ is the attention on neighbor $u \in N(v)$ when aggregating information at node u.

- Many ways to compute $\alpha_{v,u}^{(k)}$, but in the original paper

$$\alpha_{v,u}^{(k)} = \frac{exp\left(\mathbf{a}^{(k)^T}\left[\mathbf{W}^{(k)}\mathbf{h}_v^{(k)} \mid \mathbf{W}^{(k)}\mathbf{h}_u^{(k)}\right]\right)}{\sum_{u' \in N(v)} exp\left(\mathbf{a}^{(k)^T}\left[\mathbf{W}^{(k)}\mathbf{h}_v^{(k)} \mid \mathbf{W}^{(k)}\mathbf{h}_{u'}^{(k)}\right]\right)}$$

scalar attention

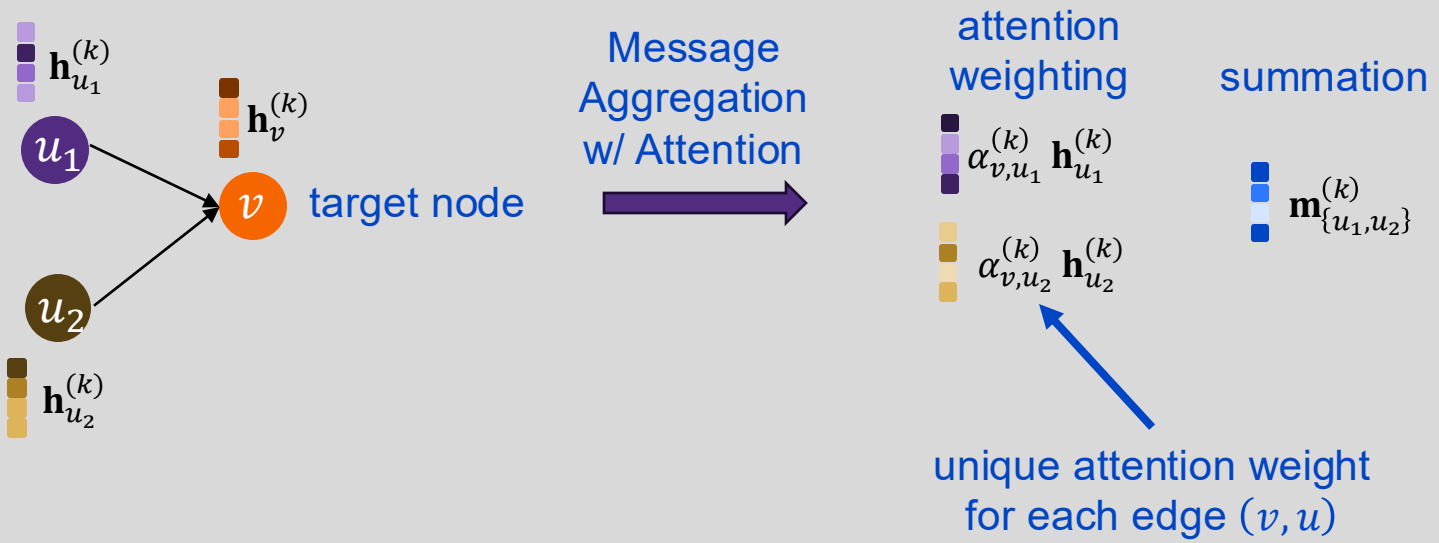
$\mathbf{a}^{(k)} \in \mathbb{R}^{2d_{k+1} \times 1}$ is a learnable attention vector

$\mathbf{W}^{(k)} \in \mathbb{R}^{d_{k+1} \times d_k}$ is a learnable weight matrix

Graph attention networks message aggregation + attention

- Formulate attention in the message passing framework

$$\mathbf{m}_{N(v)}^{(k)} = \sum_{u \in N(v)} \alpha_{v,u}^{(k)} \mathbf{h}_u^{(k)}$$



Graph attention message aggregation in matrix form

- How many attention weights, $\alpha_{v,u}^{(k)}$, do we compute at a given layer?
  - $2m$ where m is the number of edges in the graph. Why?
- We can put these in a matrix, $\widetilde{\mathbf{W}}^{(k)} \in \mathbb{R}^{|V|\times|V|}$, where

$$\widetilde{w}_{v,u}^{(k)} = \begin{cases} \alpha_{v,u}^{(k)} \; if \; (u,v) \in \mathcal{E} \\ 0 \quad if \; (u,v) \notin \mathcal{E} \end{cases}$$

- Now, we can write the message aggregation in matrix form as

$$\mathbf{M}^{(k)} = \widetilde{\mathbf{W}}^{(k)}\mathbf{H}^{(k)}$$

where $\mathbf{H}^{(k)} \in \mathbb{R}^{|\boldsymbol{V}|\times \boldsymbol{d}_k}$ so that $\mathbf{M}^{(k)} \in \mathbb{R}^{|\boldsymbol{V}|\times \boldsymbol{d}_k}$

This formulation assumes the attention weights have already been computed.

Graph attention with convolution update

- Assuming we have the message aggregation in matrix

$$\mathbf{M}^{(k)} = \widetilde{\mathbf{W}}^{(k)}\mathbf{H}^{(k)} \in \mathbb{R}^{|V|\times d_k}$$

we update the node embeddings via any differentiable update function

$$\mathbf{H}^{(k+1)} = Update^{(k)}\left(\mathbf{H}^{(k+1)}, \mathbf{M}^{(k)}\right)$$

- The typical update function applies the same approach as graph convolution networks

$$\mathbf{H}^{(k+1)} = \mathbf{M}^{(k)}\mathbf{W}^{(k)^T} \in \mathbb{R}^{|V|\times d_{k+1}}$$

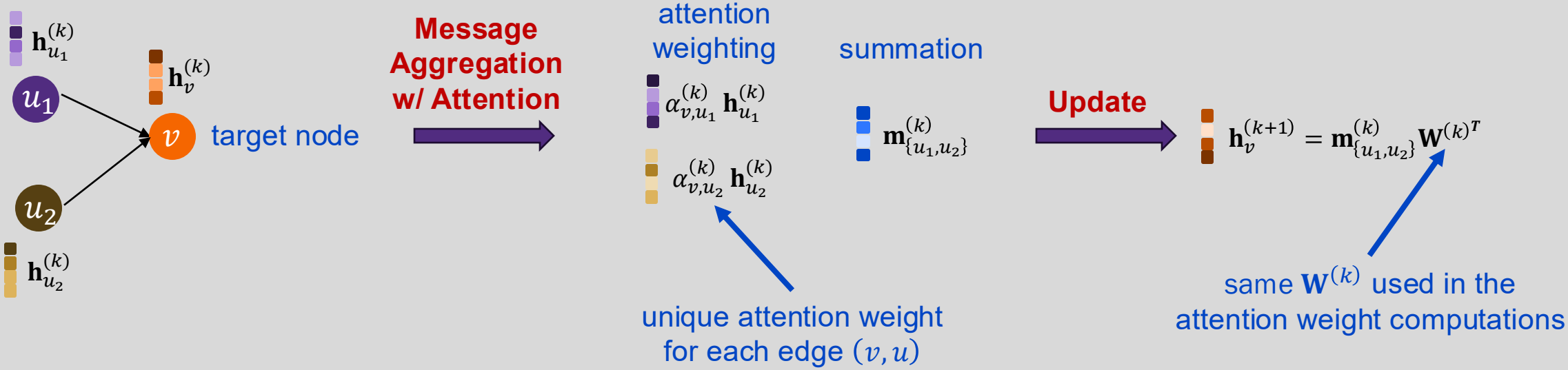
This is the same $\mathbf{W}^{(k)}$ used in the attention weight computations

Graph attention with convolution update

- Formulate attention in the message passing framework

$$\mathbf{m}_{N(v)}^{(k)} = \sum_{u \in N(v)} \alpha_{v,u}^{(k)} \mathbf{h}_u^{(k)}$$

$$\mathbf{h}^{(k+1)} = \mathbf{m}_{N(v)}^{(k)} \mathbf{W}^{(k)T}$$



Graph attention with convolution update layer

Putting it all together, graph attention with convolution update involves the following steps at each layer:

1. Compute the attention weights

$$\alpha_{v,u}^{(k)} = \frac{exp\left(\mathbf{a}^{(k)^T}\left[\mathbf{W}^{(k)}\mathbf{h}_v^{(k)} \mid \mathbf{W}^{(k)}\mathbf{h}_u^{(k)}\right]\right)}{\sum_{u' \in N(v)} exp\left(\mathbf{a}^{(k)^T}\left[\mathbf{W}^{(k)}\mathbf{h}_v^{(k)} \mid \mathbf{W}^{(k)}\mathbf{h}_{u'}^{(k)}\right]\right)}$$

2. Aggregate node messages with attention weighting

$$\mathbf{m}_{N(v)}^{(k)} = \sum_{u \in N(v)} \alpha_{v,u}^{(k)}\mathbf{h}_u^{(k)} \rightarrow \mathbf{M}^{(k)} = \widetilde{\mathbf{W}}^{(k)}\mathbf{H}^{(k)}$$

3. Update node embeddings

$$\mathbf{h}_v^{(k+1)} = \mathbf{m}_{N(v)}^{(k)}\mathbf{W}^{(k)^T} \rightarrow \mathbf{H}^{(k+1)} = \mathbf{M}^{(k)}\mathbf{W}^{(k)^T} \in \mathbb{R}^{|V| \times \boldsymbol{d_{k+1}}}$$

Multi-head graph attention

- Use H different *attention heads* to aggregate messages
- Results in H different messages, one for each attention head, $h \in [1, H]$
- Stabilizes attention mechanism learning

$$\mathbf{m}_{N(v)}^{(k,h)} = \sum_{u \in N(v)} \alpha_{v,u}^{(k,h)} \mathbf{h}_u^{(k)}$$

where

$$\alpha_{v,u}^{(k)} = \frac{exp\left(\mathbf{a}^{(k,h)^T}\left[\mathbf{W}^{(k,h)}\mathbf{h}_v^{(k)} \mid \mathbf{W}^{(k,h)}\mathbf{h}_u^{(k)}\right]\right)}{\sum_{u' \in N(v)} exp\left(\mathbf{a}^{(k,h)^T}\left[\mathbf{W}^{(k,h)}\mathbf{h}_v^{(k)} \mid \mathbf{W}^{(k,h)}\mathbf{h}_{u'}^{(k)}\right]\right)}$$

$\mathbf{a}^{(k,h)} \in \mathbb{R}^{2d_{k+1} \times 1}$

different attention vector for each attention head

$\mathbf{W}^{(k,h)} \in \mathbb{R}^{d_{k+1} \times d_k}$

different weight matrix for each attention head

Multi-head graph attention with convolution update

- Aggregated message from each attention head is updated individually
- Updates are aggregated (e.g., concatenation or summation) to form new embedding for node v

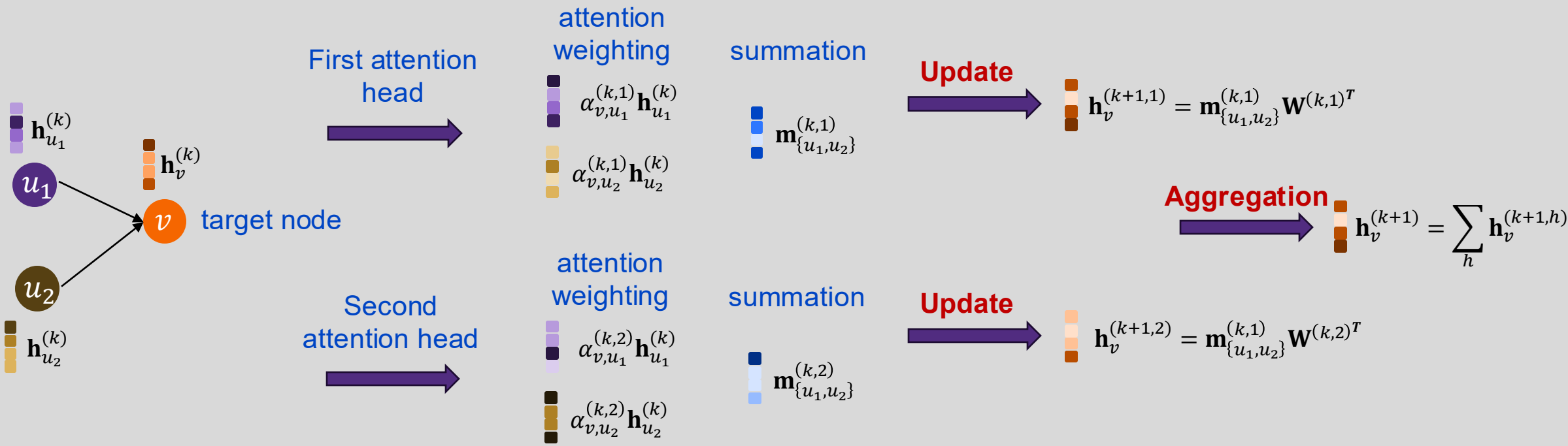
$$\mathbf{m}_{N(v)}^{(k,h)} = \sum_{u \in N(v)} \alpha_{v,u}^{(k,h)} \mathbf{h}_u^{(k)}$$

$$\mathbf{h}^{(k+1,h)} = \mathbf{m}_{N(v)}^{(k,h)} \mathbf{W}^{(k,h)^T}$$

$$\mathbf{h}^{(k+1)} = \sum_h \mathbf{h}^{(k+1,h)}$$

Multi-head graph attention with convolution update

- Here, assume we have two attention heads



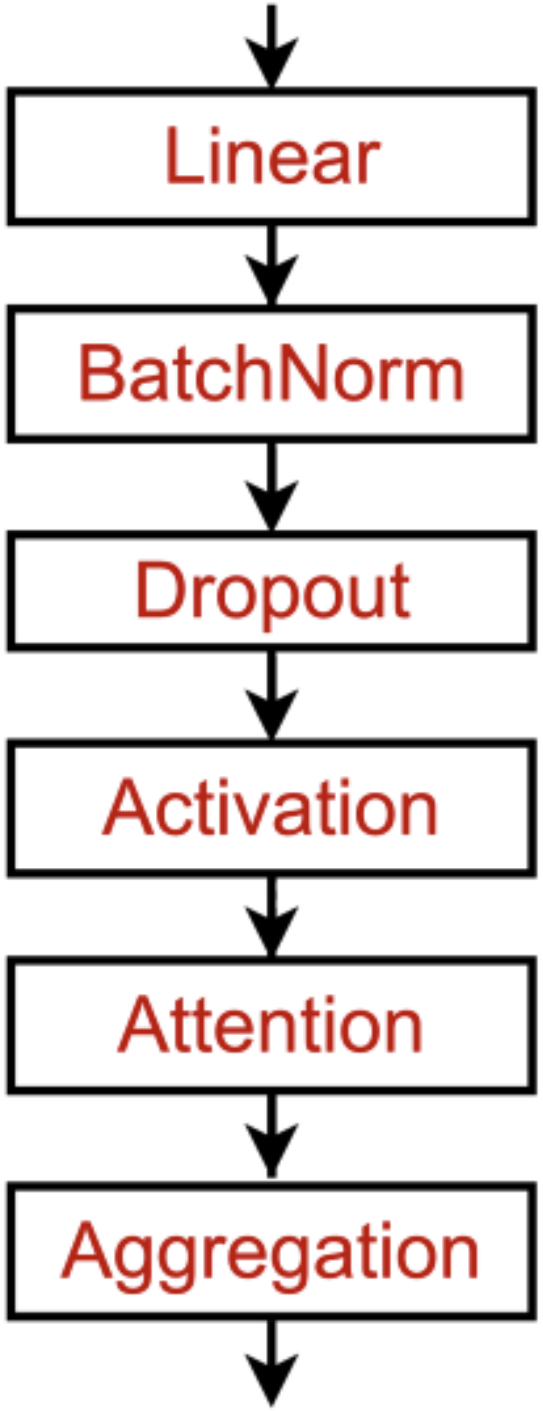
Benefits of Attention

- **Key benefit:** Allows for implicitly specifying different importance values for each neighbor node
- Computational efficiency
  - Attention weights can be computed in parallel across all graph edges
  - Aggregation can by parallelized across all nodes
- Storage efficient
  - Sparse matrix operations require $\leq O(V + \mathcal{E})$ storage
  - Fixed number of model parameters independent of graph size (node count)
- A note about implementation:
  - The message passing framework is useful for conceptual understanding of GNNs
  - In practice, many of the matrix operations are implemented in ways that mix aspects of neighbor node aggregation and node update for efficiency

Applicable deep learning tools

A GNN layer in practice

- We often want to use familiar deep learning modules to create our GNN layers
  - Batch normalization – stabilize network training
  - Dropout – prevent overfitting
  - Many other useful deep learning modules



Batch normalization

- Goal: stabilize network training
- Approach: Given a batch of N inputs (e.g., node embeddings)
  - Recenter the embeddings to zero mean
  - Rescale the embeddings to unit variance

Input: $\mathbf{X} \in \mathbb{R}^{N \times d}$

Learnable Parameters: $\boldsymbol{\gamma}, \boldsymbol{\beta} \in \mathbb{R}^{d}$

Output: $\mathbf{Y} \in \mathbb{R}^{N \times d}$

Step 1:
Compute the mean and variance for each latent feature (embedding column j) over the N embeddings

$$u_j = \frac{1}{N}\sum_{i=1}^{N} x_{i,j}$$

$$\sigma_j^2 = \frac{1}{N}\sum_{i=1}^{N}\left(x_{i,j} - \mu_j\right)^2$$

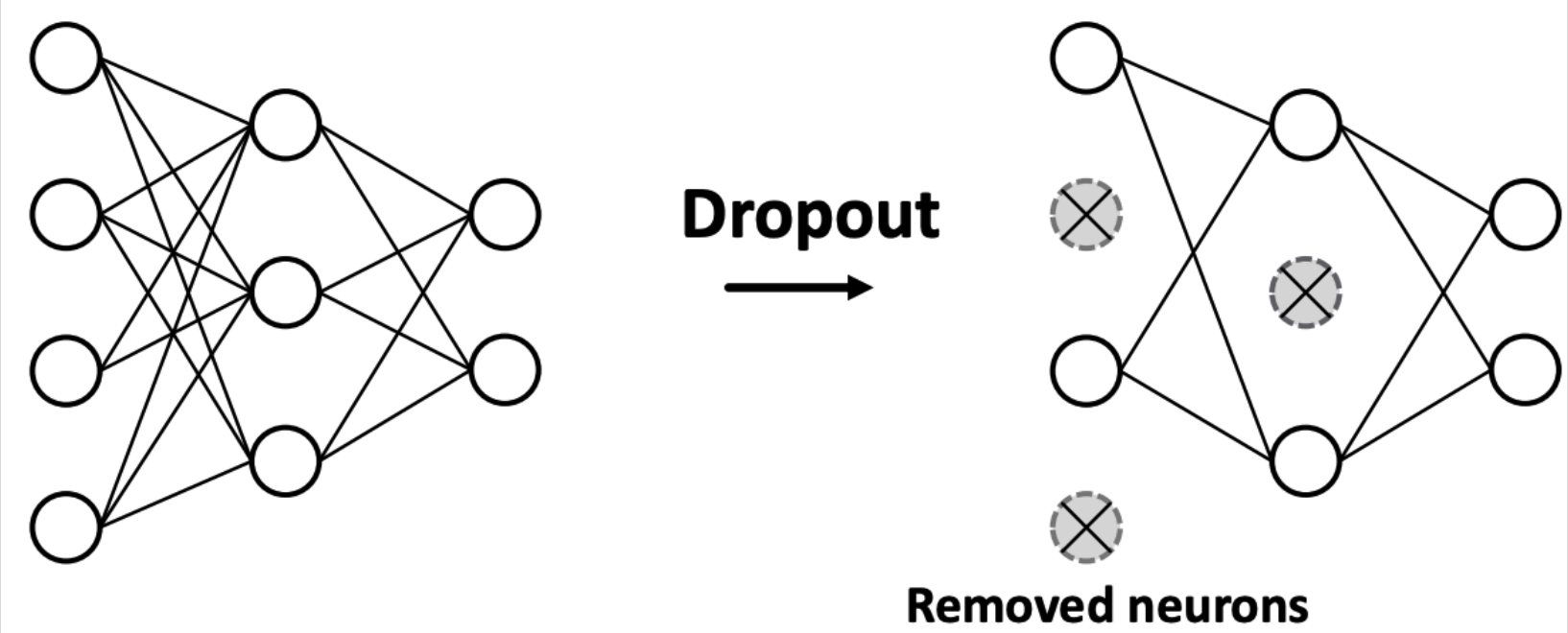
Step 2:
Normalize the embedding feature

$$\hat{x}_{i,j} = \frac{x_{ij} - \mu_j}{\sqrt{\sigma_j^2 + \epsilon}}$$

$$y_{i,j} = \gamma_j \hat{x}_{i,j} + \beta_j$$

Dropout regularization in standard neural networks

- Regularize a deep neural network to prevent overfitting
- Approach
  - During training: with some probability p, randomly turn off network neurons (i.e., set output to zero)
  - During test / deployment: use all neurons



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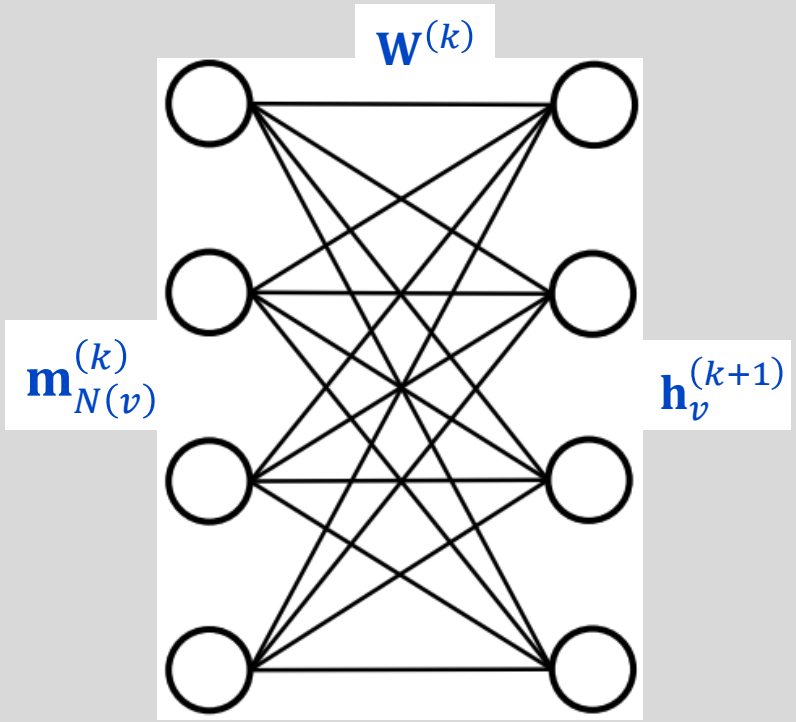
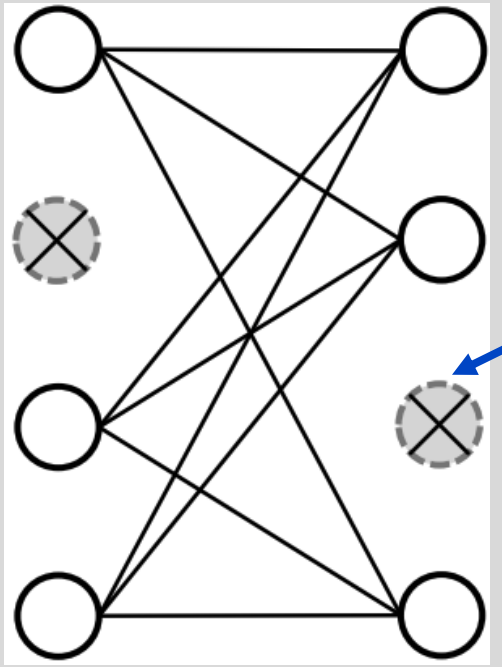
Dropout regularization in GNNs

- In a GNN, dropout is applied in the *update* function
- Example, in GCN the update for node v is $\mathbf{h}_v^{(k+1)} = \sigma\left(\mathbf{W}^{(k)}\mathbf{m}_{N(v)}^{(k)}\right)$
  - Dropout implemented by removing neurons in $\mathbf{W}^{(k)}$



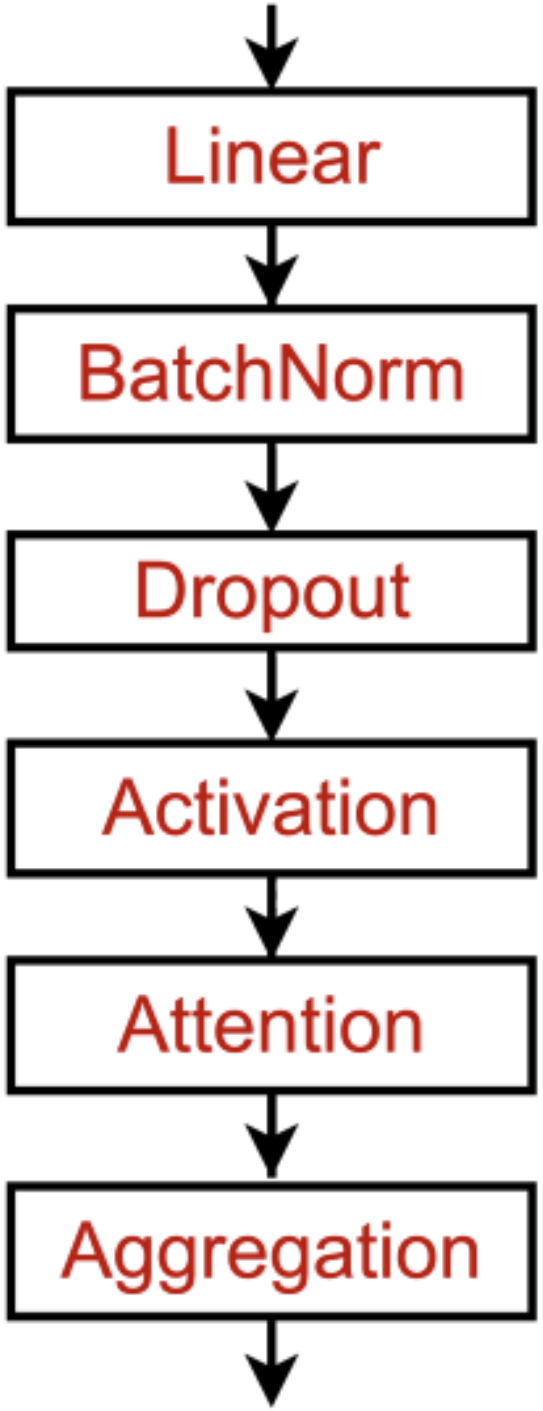


Dropout removes latent features from the aggregated message, NOT nodes in the graph

Image credit: Jure Leskovec, http://cs224w.stanford.edu

GNN layers in practice

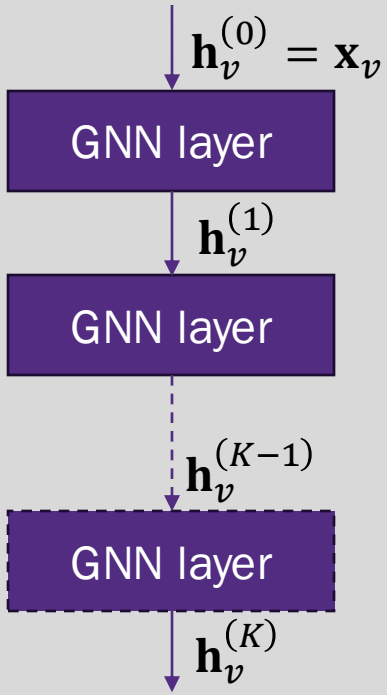
- Modern deep learning modules are useful in developing practical GNN layers
- Designing novel GNN layers is an active research domain



Stacking GNN layers

Stacking GNN layers

- How do we construct a GNN with multiple layers?
- Standard approach is to stack layers sequentially
- Initial input: $\mathbf{h}_v^{(0)} = \mathbf{x}_v$ is the input feature vector for v
- Output: Node embedding $\mathbf{h}_v^{(K)}$ after K layers

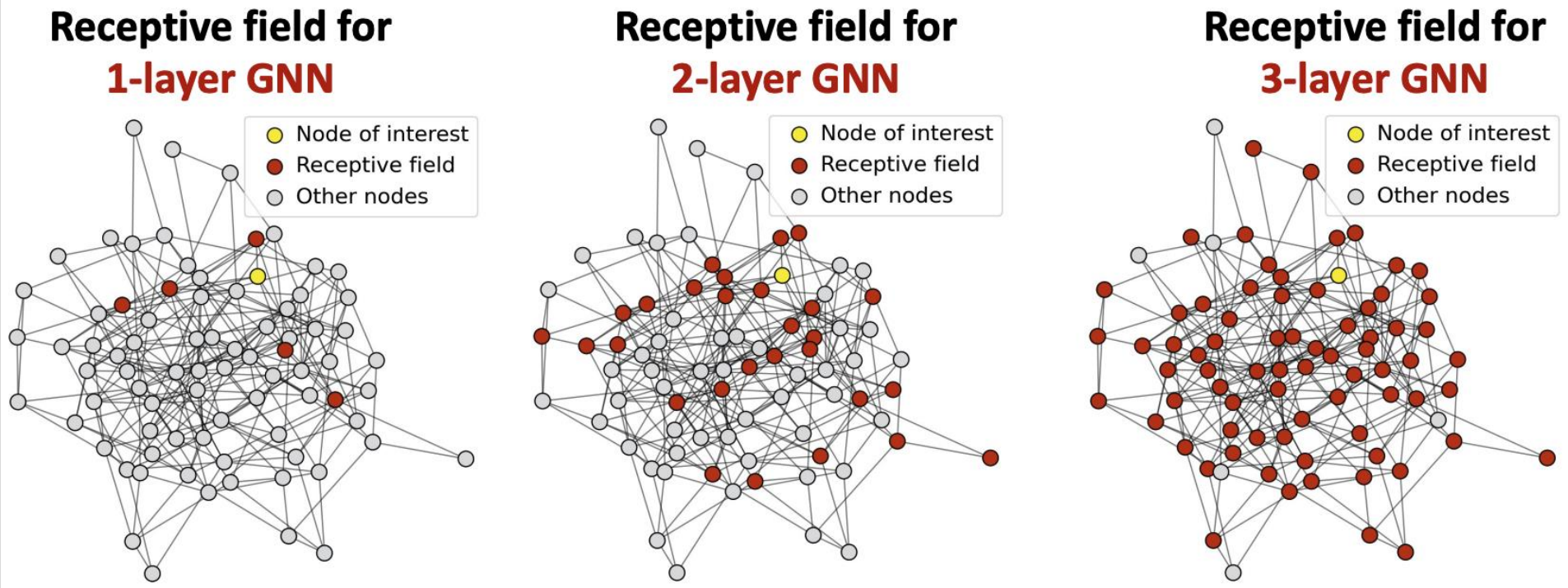


Over-smoothing in deep GNNs

- Stacking many GNN layers can cause over-smoothing
- Over-smoothing occurs when all node embeddings converge to the same value – i.e., all embedding vectors are the same
- **Why does this happen?**

Receptive field of a GNN

- Receptive field: the set of nodes that contribute to the aggregated neighborhood message for a node of interest
- In a K-layer GNN, each node has a receptive field of its K-hop neighborhood



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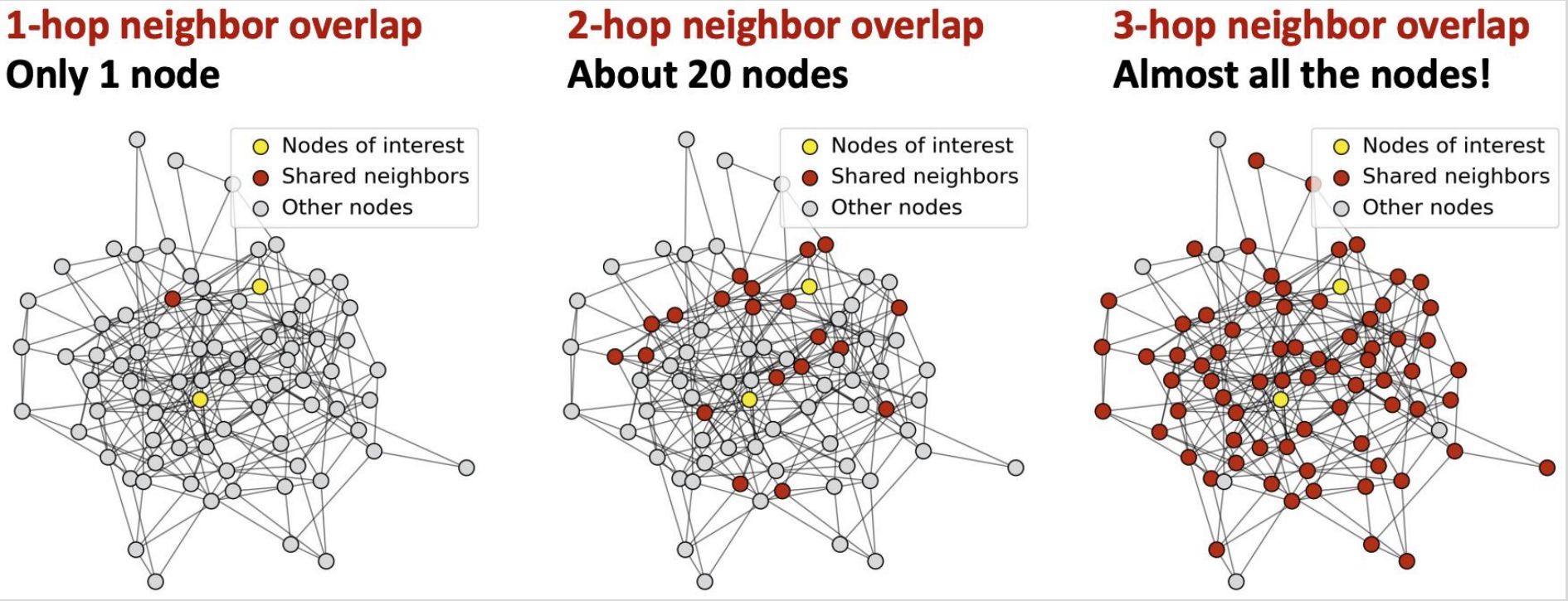
Receptive field of a GNN

- Receptive field overlap for two nodes
- The shared neighbors quickly grows when increasing the field hop size (i.e., number of GNN layers)
- Results in very similar messages being input to each node's update function, yielding similar embeddings for both



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Addressing over-smoothing

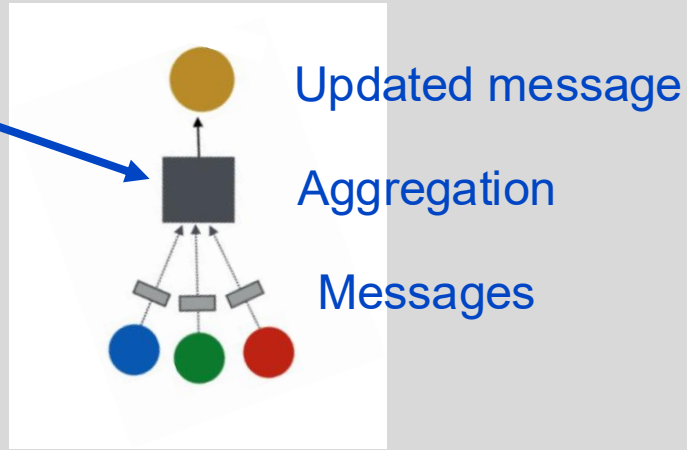
- Be conservative with adding GNN layers – adding too many layers will cause over-smoothing
- Analyze the receptive field necessary for your problem (e.g., by computing the network diameter – longest shortest path)
- Set the number of GNN layers to be slightly more (say plus 1-3) than the estimated receptive field requirement

How can we enhance the expressive power of a GNN if the receptive field requirement limits the model to a small number of GNN layers?

Increasing expressive power for GNNs

- Increase the expressive power within the GNN layer
- In the GNN layers we discussed (GCN, GraphSAGE, GAT), the aggregations and transformations only include one linear layer
- We can add layers to these to form a deeper network with the layer – this does not change the receptive field

The aggregation and updates can use an MLP



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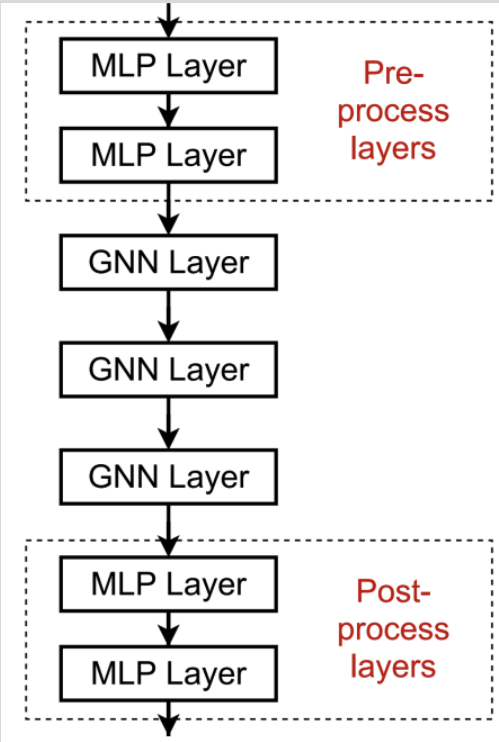
Increasing expressive power for GNNs

- Add network layers that do not pass messages
- The overall network can contain non-message passing (GNN) layers
  - Example: add MLP layers before and after GNN layers to pre/post process next/prior layer output



Pre-processing: important when encoding features is necessary (e.g., when nodes represent images or text)

Post-processing: important when reasoning / transformation over node embeddings is required (e.g., graph classification)

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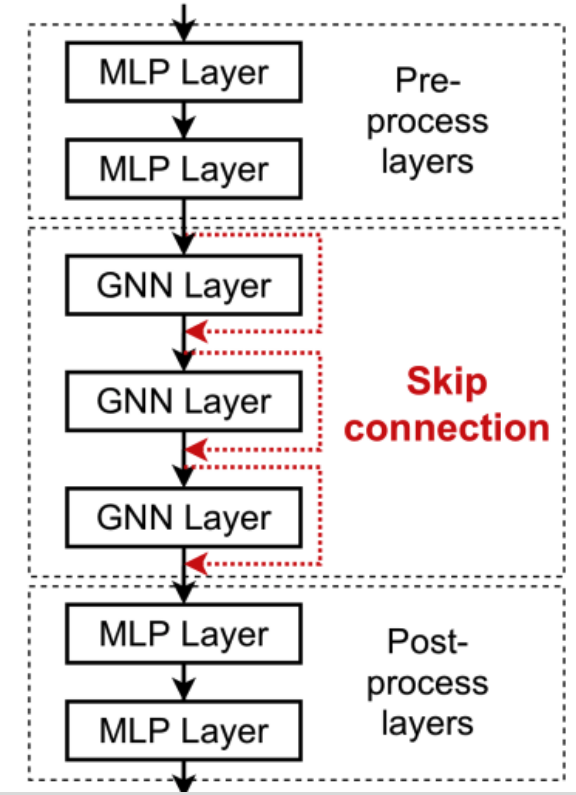
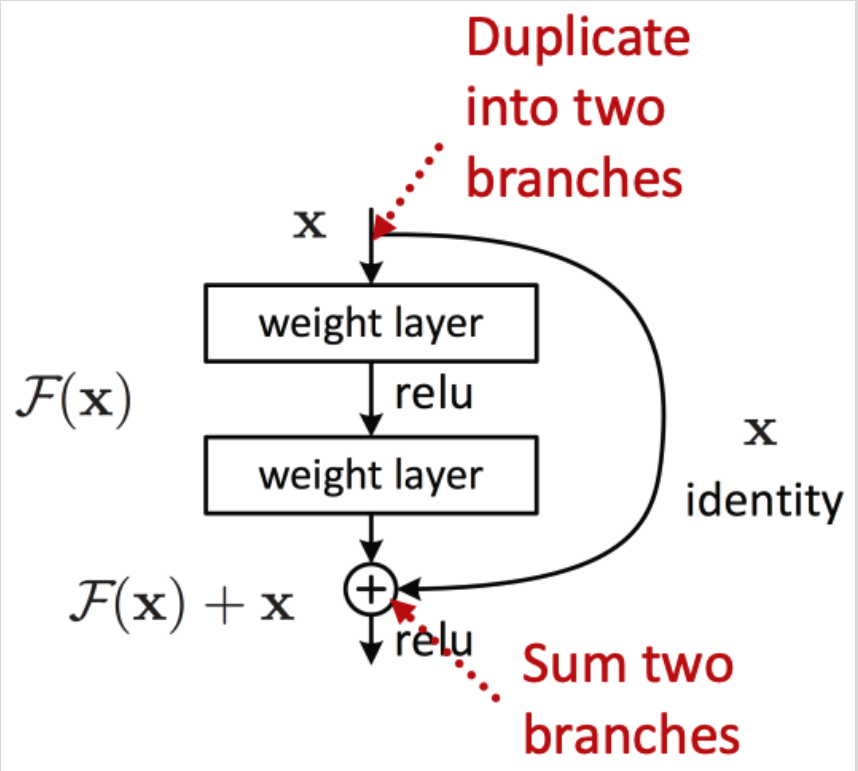
But I really need more GNN layers! Skip connections

- Skip connections can be used to directly connect earlier GNN layers to deeper layers
- The node embeddings from earlier layers may prevent over-smoothing
- Acts like a mixture of shallow GNNs and deep GNNs





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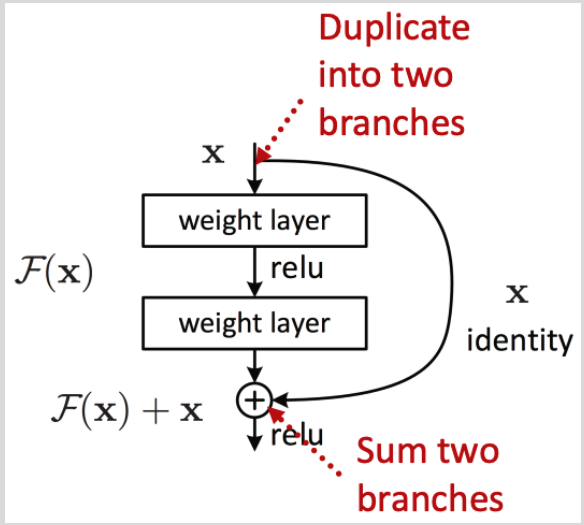
GCN with skip connections

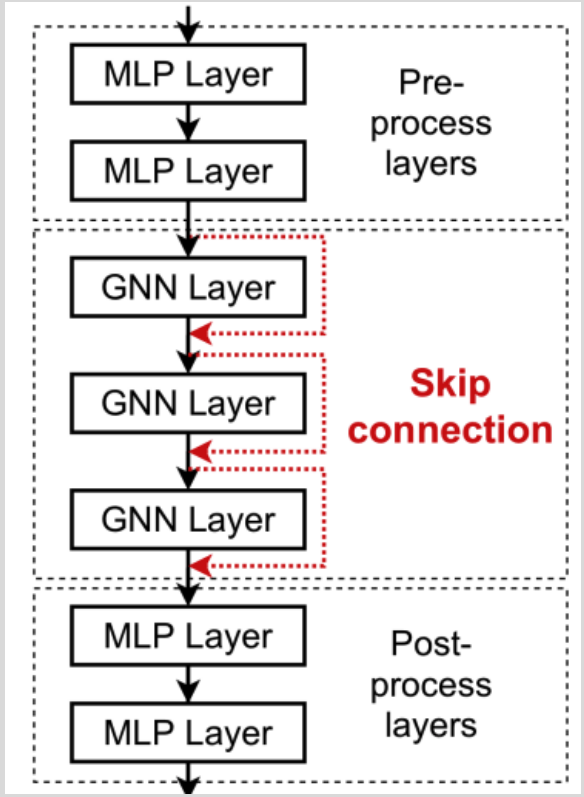
- Standard GCN layer

$$\mathbf{h}_v^{(k+1)} = \sigma\left(\underbrace{\mathbf{W}^{(k)} \sum_{u \in N(v) \cup \{v\}} \frac{\mathbf{h}_u^{(k)}}{|N(v)|}}_{f(x)}\right)$$

- GCN layer with skip connection from layer $k - j$ (typically $j = 1$)

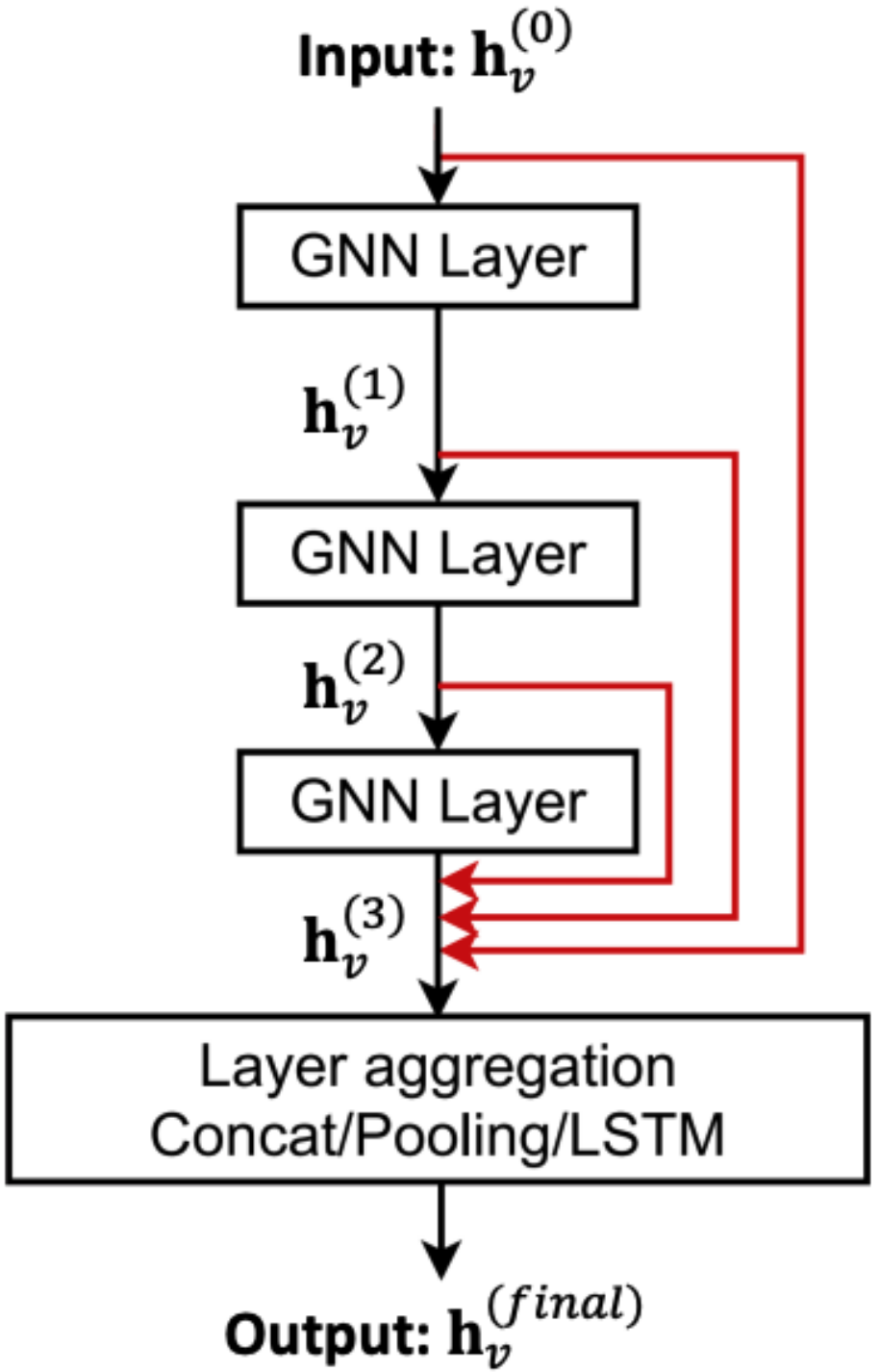
$$\mathbf{h}_v^{(k+1)} = \sigma\left(\underbrace{\mathbf{h}_v^{(k-j)}}_{x} + \underbrace{\mathbf{W}^{(k)} \sum_{u \in N(v) \cup \{v\}} \frac{\mathbf{h}_u^{(k)}}{|N(v)|}}_{f(x)}\right)$$





Other skip connection options

- Directly skip to the last layer
- The final layer aggregates from all the GNN layer node embeddings in the previous layers



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Graph manipulations

Why manipulate graphs

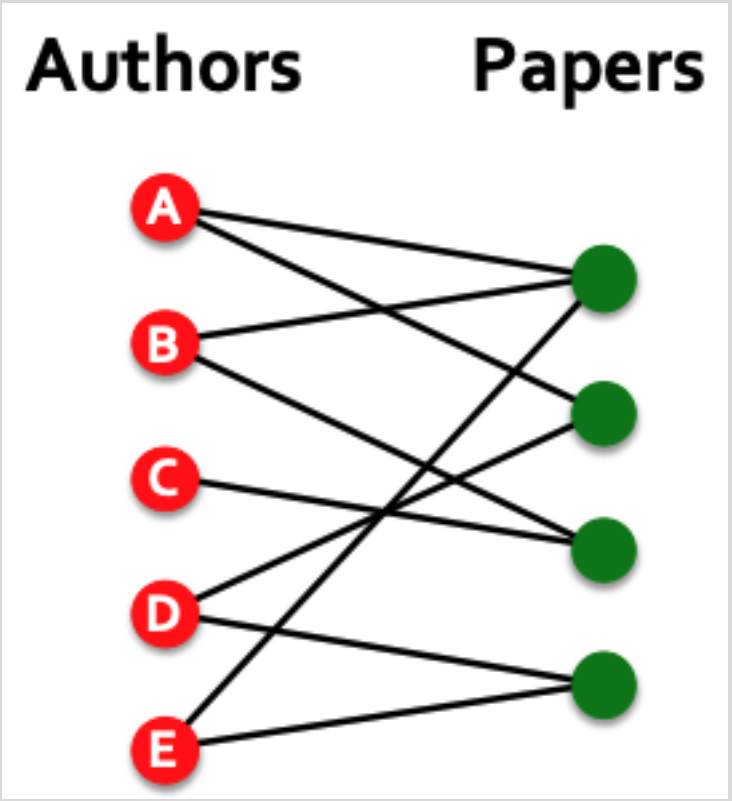
- We've assumed that the network graph = the GNN structure
- We might want to break this assumption if:
  - The input graph lacks features → no messages
  - The graph is too sparse → inefficient message passing
  - The graph is too dense → message passing computation is expensive
  - The graph is too large → cannot fit the computation graph into a GPU

Why manipulate graphs

- We've assumed that the network graph = the GNN structure
- We might want to break this assumption if:
  - The input graph lacks features → apply feature augmentation (later lecture)
  - The graph is too sparse → add virtual nodes
  - The graph is too dense → apply GraphSAGE or other neighbor sampling
  - The graph is too large → sample subgraphs to compute embeddings (later lecture, time permitting)

Adding virtual edges

- Goal is to augment a sparse graph
- Common approach is to connect 2-hop neighbors via virtual edges
- Mathematically, equivalent to using $\mathbf{A} + \mathbf{A}^2$ instead of the adjacency matrix
- Common use case is bipartite graphs
  - Author-to-papers (they authored)
  - 2-hop virutal edges make an author-author collaboration graph



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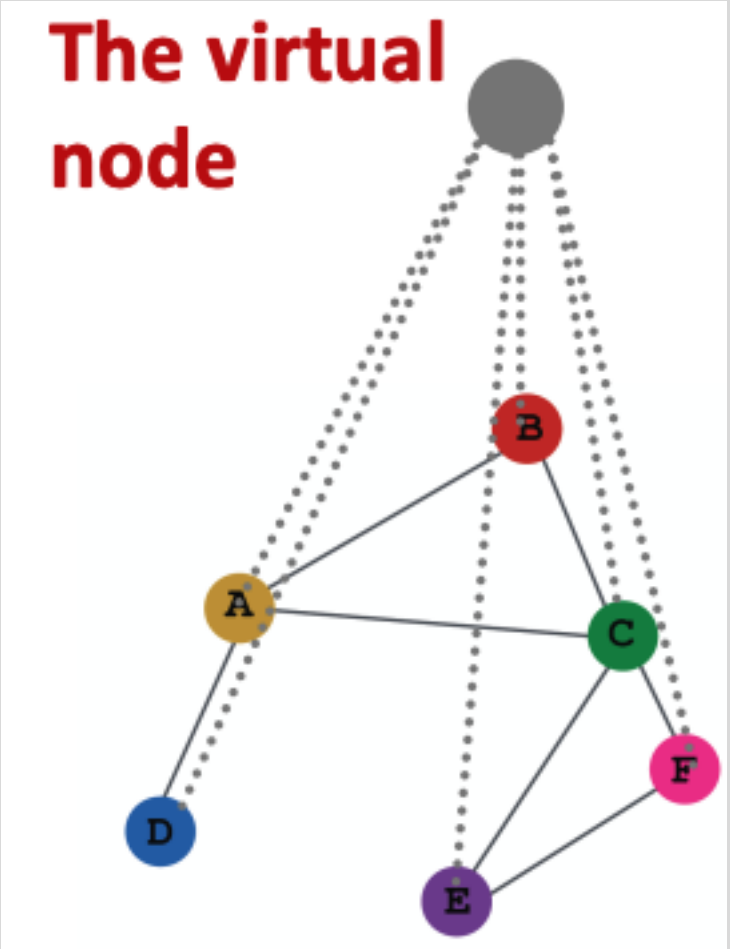
Adding virtual nodes

- Goal is to augment a sparse graph
- Virtual node will connect to all nodes in the graph (or subgraph)
  - Suppose in a sparse graph, two nodes have a shortest path distance > 2
  - After adding the virtual node, all nodes have a shortest path distance of 2
    - node U – virtual node – node V
  - Greatly improves message passing



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